問1

$$h(t) = E[\{(X-\mu_X) - t(Y-\mu_Y)\}^2]$$

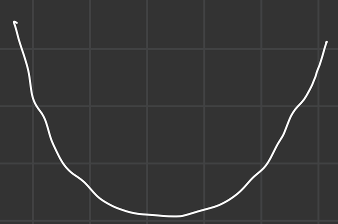
を用いて

$$\{E[(X-\mu_X)(Y-\mu_Y)]\}^2 \leqq E[(X-\mu_X)^2]E[(Y-\mu_Y)^2]$$

$$h(t)$$
$$= E[(X-\mu_X)^2 + t^2(Y-\mu_Y)^2 - 2t(X-\mu_X)(Y-\mu_Y)]$$
$$= E[(X-\mu_X)^2] + t^2E[(Y-\mu_Y)^2] - 2tE[(X-\mu_X)(Y-\mu_Y)]$$

$$h'(t) = 2tE[(Y-\mu_Y)^2] - 2E[(X-\mu_X)(Y-\mu_Y)]$$

	$\frac{E[(X-\mu_X)(Y-\mu_X)]}{E[(Y-\mu_Y)^2]}$	
$-$	0	$+$
$+$	$+$	$+$
↘		↗



$$h\left(\frac{E[(X-\mu_X)(Y-\mu_Y)]}{E[(Y-\mu_Y)^2]}\right)$$

$$= E[(X-\mu_X)^2] + \frac{E[(X-\mu_X)(Y-\mu_Y)]^2}{E[(Y-\mu_Y)^2]} - \frac{2E[(X-\mu_X)(Y-\mu_Y)]^2}{E[(Y-\mu_Y)^2]}$$

$$= \frac{E[(X-\mu_X)^2]E[(Y-\mu_Y)^2] - E[(X-\mu_X)(Y-\mu_Y)]^2}{E[(Y-\mu_Y)^2]}$$

$$Var(X) - \frac{Cov(X,Y)^2}{Var(Y)} = 0$$

$$Cov(X,Y) = \pm\sqrt{Var(X)}\sqrt{Var(Y)}$$

$h(t) \geqq 0$ であり

$t = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{E[(Y-\mu_Y)^2]}$ のとき

問2

$X \sim f(x)$

$Y \sim g(y)$

$E[X] = E[Y] = \mu$

$Var[X] = Var[Y] = \sigma^2$

$Corr(X,Y) = \rho$

(1)

$W \sim Ber(p)$

$Y = Y$

$Z = WX + (1-W)Y$

$$F_Z(z) = P(Z \leq z)$$
$$= P(WX + (1-W)Y \leq z)$$
$$= P(W=0, Y \leq z) + P(W=1, X \leq z)$$
$$= (1-p) F_Y(z) + p F_X(z)$$

$$f_Z(z) = p f_X(z) + (1-p) f_Y(z)$$ //

$$E[Z] = \int z f_Z(z)\, dz$$
$$= \int z \left(p f_X(z) + (1-p) f_Y(z)\right) dz$$
$$= p \int z f_X(z) dz + (1-p) \int z f_Y(z) dz$$
$$= p E[X] + (1-p) E[Y]$$
$$= \mu$$ //

$$E[Z^2]$$
$$= \int z^2 f_Z(z) dz$$
$$= \int z^2 \left(p f_X(z) + (1-p) f_Y(z)\right) dz$$
$$= p \int z^2 f_X(z) dz + (1-p) \int z^2 f_Y(z) dz$$
$$= p E[X^2] + (1-p) E[Y^2]$$

$$Var(Z) = E[Z^2] - E[Z]^2$$
$$= p E[X^2] + (1-p) E[Y^2] - \mu^2$$
$$= \sigma^2$$ //

$$Var(X) = E[X^2] - \mu^2$$

$$\therefore \quad E[X^2] = \sigma^2 + \mu^2$$
$$E[Y^2] = \sigma^2 + \mu^2$$

(2)

$$Var(U) = Var(wX + (1-w)Y)$$
$$= w^2 Var(X) + (1-w)^2 Var(Y) + 2w(1-w) Cov(X,Y)$$
$$= w^2\sigma^2 + (1 - 2w + w^2)\sigma^2 + (2w - 2w^2)\rho\sigma^2$$
$$= 2(1-\rho)\sigma^2 w^2 - 2\sigma^2 w + 2\rho\sigma^2 w + \sigma^2$$

$$\frac{\partial Var(U)}{\partial w} = 4(1-\rho)\sigma^2 w - 2\sigma^2 + 2\rho\sigma^2$$
$$= 2(1-\rho)\sigma^2 (2w - 1)$$

$$\therefore \quad w = \frac{1}{2}$$

問) 3

(1)

$$E[X(Y-E[Y|X])] - E[X]\cdot E[Y-E[Y|X]]$$

$$= E[XY - XE[Y|X]] - E[X](E[Y]-E[Y])$$

$$= E[XY - XE[Y|X]]$$

$$= E[XY] - E[XE[Y|X]]$$

$$E[XE[Y|X]] = \int x\, E[Y|X]\, f_X(x)\,dx$$

$$= \int x\left(\int y\cdot f_{Y|X}(y|x)\,dy\right) f_X(x)\,dx$$

$$= \int x\left(\int y\cdot \frac{f_{X,Y}(x,y)}{f_X(x)}\,dy\right) f_X(x)\,dx$$

$$= \int x \int y\, f_{X,Y}(x,y)\,dy\,dx$$

$$= E[XY]$$

よって

$$Cov(X, Y-E[Y|X]) = 0$$

(2)

$$Var(Y-E[Y|X]) = E[Var(Y|X)]$$

$$Var(Y-E[Y|X])$$

$$= Var(Y) + Var(E[Y|X])$$

$$= Var(Y) + E(E[Y|X]^2) - E(E[Y|X])^2$$

$$E[Var(Y|X)]$$

$$= E[\ E[(Y|X)^2] - E[Y|X]^2]$$

$$Var(X) = E[Var(X|Y)] + Var(E[X|Y])$$

$$\rightarrow\ Var(Y) = E[Var(Y|X)] + Var(E[Y|X])$$

$\therefore$

$$E[Var(Y|X)] = Var(Y) - Var(E[Y|X])$$

(2) $Var(Y - E[Y|X]) = E[Var(Y|X)]$ を示す.

$Var(Y - E[Y|X])$

$$= E[\{(Y - E[Y|X]) - E[Y - E[Y|X]]\}^2]$$

$$= E[(Y - E[Y|X] - E[Y] - E[E[Y|X]])^2]$$

$$= E[(Y - E[Y|X])^2]$$

$$= E[Y^2 - 2YE[Y|X] + E^2[Y|X]]$$

$$= E[Y^2] - 2E[YE[Y|X]] + E[E[Y|X]^2]$$

$$= E[E[Y^2|X]] - 2E[Y \cdot E[Y|X]] + E[E[Y|X]^2]$$

$$= E[E[Y^2|X]] - 2E[E[Y \cdot E[Y|X]|X]] + E[E[Y|X]^2]$$

$$= E[E[Y^2|X]] - 2E[E[Y|X] \cdot E[Y|X]] + E[E[Y|X]^2]$$

$$= E[E[Y^2|X] - E[Y|X]^2]$$

$$= E[Var(Y|X)]$$

問4

$$f_{X_1|X_2} = \frac{f_{X_1,X_2}}{f_{X_2}}$$

$$= \frac{\frac{1}{2\pi}\frac{1}{\sqrt{1-\rho^2}\sigma_1\sigma_2}\exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2-2\rho\frac{x_1-\mu_1}{\sigma_1}\frac{x_2-\mu_2}{\sigma_2}+\left(\frac{x_2-\mu_2}{\sigma_2}\right)^2\right]\right\}}{\frac{1}{\sqrt{2\pi}\sigma_2}\exp\left\{-\frac{(x_2-\mu_2)^2}{2\sigma_2^2}\right\}}$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1}\cdot\frac{1}{\sqrt{1-\rho^2}}$$

$$\cdot\exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2-2\rho\cdot\frac{x_1-\mu_1}{\sigma_1}\cdot\frac{x_2-\mu_2}{\sigma_2}+\rho^2\left(\frac{x_2-\mu_2}{\sigma_2}\right)^2\right]\right\}$$

$$= \frac{1}{\sqrt{2\pi}\left(\sqrt{(1-\rho^2)\sigma_1^2}\right)}$$

$$\exp\left\{-\frac{\left\{x_1-\left(\mu_1+\frac{\rho\sigma_1}{\sigma_2}(x_2-\mu_2)\right)\right\}^2}{2(1-\rho^2)\sigma_1^2}\right\}$$

$$= N\left(\mu_1+\frac{\rho\sigma_1}{\sigma_2}(x_2-\mu_2), (1-\rho^2)\cdot\sigma_1^2\right)$$

無相関 ⇒ 独立

$\rho=0$ のとき

$f_X(x,\mu,\Sigma)$

$$= \frac{1}{2\pi}\cdot\frac{1}{\sigma_1\sigma_2}\exp\left\{-\frac{1}{2}\left[\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2+\left(\frac{x_2-\mu_2}{\sigma_2}\right)^2\right]\right\}$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1}\exp\left\{-\frac{(x_1-\mu_1)^2}{2\sigma_1^2}\right\}\times\frac{1}{\sqrt{2\pi}\sigma_2}\exp\left\{-\frac{(x_2-\mu_2)^2}{2\sigma_2^2}\right\}$$

$$= f_{X_1}(x_1|\mu_1,\sigma_1)\cdot f_{X_2}(x_2|\mu_2,\sigma_2)$$

よって独立。

独立のとき

$$\mathrm{cov}(X_1,X_2) = E[X_1X_2]-E[X_1][X_2]$$

$$= \int_{-\infty}^{\infty} x_1x_2\, f_X(x|\mu,\Sigma)\,dx$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_1x_2\, f_{X_1}(x_1|\mu_1,\sigma_1)\, f_{X_2}(x_2|\mu_2,\sigma_2)\,dx_1dx_2$$

$$- E[X_1]E[X_2]$$

$$= \int_{-\infty}^{\infty} x_1 f_{X_1}(x_1|\mu_1,\sigma_1)\,dx_1\cdot\int_{-\infty}^{\infty} x_2 f_{X_2}(x_2|\mu_2,\sigma_2)\,dx_2$$

$$- E[X_1]E[X_2]$$

$$= E[X_1]E[X_2]-E[X_1]E[X_2]$$

$$= 0$$

$\therefore\ \rho=0$ より無相関

問5

$$X \sim N(\mu_1, \sigma_1^2)$$

$$Y \sim N(\mu_2, \sigma_2^2)$$

① 畳み込み

$$T = Y \qquad X = S - T$$
$$S = X + T \qquad Y = T$$

$$J((S,T) \to (x,y)) = \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} = 1$$

$$f_{S,T}(s,t) = f_X(s-t)\, f_Y(t)$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1} \exp\left\{-\frac{(s-t-\mu_1)^2}{2\sigma_1^2}\right\} \cdot \frac{1}{\sqrt{2\pi}\sigma_2} \exp\left\{-\frac{(t-\mu_2)^2}{2\sigma_2^2}\right\}$$

$$= \frac{1}{2\pi\sigma_1\sigma_2} \exp\left\{-\frac{1}{2}\left[\left(\frac{s-t-\mu_1}{\sigma_1}\right)^2 + \left(\frac{t-\mu_2}{\sigma_2}\right)^2\right]\right\}$$

$$f_S(s) = \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_1\sigma_2} \exp\left\{-\frac{1}{2}\left[\left(\frac{s-t-\mu_1}{\sigma_1}\right)^2 + \left(\frac{t-\mu_2}{\sigma_2}\right)^2\right]\right\} dt$$

$$\frac{(s-\mu_1)^2}{(\sigma_1^2+\sigma_2^2)} + \frac{\mu_2^2}{(\sigma_1^2+\sigma_2^2)} + \frac{2\mu_2(s-\mu_1)}{\sigma_1^2+\sigma_2^2}$$

$$= \frac{((s-\mu_1)+\mu_2)^2}{\sigma_1^2+\sigma_2^2}$$

② 積率母関数

$$Z = X + Y$$

$$\varphi_Z(t) = E[e^{it(X+Y)}]$$
$$= E[e^{itX} \cdot e^{itY}]$$
$$= E[e^{itX}] \cdot E[e^{itY}]$$
$$= \varphi_X(t) \cdot \varphi_Y(t)$$

$$\varphi_X(t) = \exp\left\{\mu_1 it - \frac{\sigma_1^2 t^2}{2}\right\}$$

$$\varphi_Y(t) = \exp\left\{\mu_2 it - \frac{\sigma_2^2 t^2}{2}\right\}$$

$$\varphi_Z(t) = \exp\left\{(\mu_1+\mu_2)it - \frac{(\sigma_1^2+\sigma_2^2)t^2}{2}\right\}$$

$$\therefore f_Z(z) = N(\mu_1+\mu_2,\ \sigma_1^2+\sigma_2^2)$$

$$\frac{1}{\sqrt{2\pi}\sqrt{\sigma_1^2+\sigma_2^2}} \exp\left\{-\frac{(z-(\mu_1+\mu_2))^2}{2(\sigma_1^2+\sigma_2^2)}\right\}$$

$$-\frac{1}{2}$$

$$\frac{1}{\sigma_1^2}(t-(s-\mu_1))^2 + \frac{1}{\sigma_2^2}(t-\mu_2)^2$$

$$= \frac{1}{\sigma_1^2}(t^2 - 2(s-\mu_1)t + (s-\mu_1)^2) + \frac{1}{\sigma_2^2}(t^2 - 2\mu_2 t + \mu_2^2)$$

$$= \left(\frac{\sigma_1^2+\sigma_2^2}{\sigma_1^2\sigma_2^2}\right)\left(t - \frac{\sigma_2^2(s-\mu_1)+\sigma_1^2\mu_2}{\sigma_1^2+\sigma_2^2}\right)^2 + \frac{((s-\mu_1)-\mu_2)^2}{\sigma_1^2+\sigma_2^2}$$

$$\frac{1}{\sigma_1^2\sigma_2^2} \cdot \frac{1}{\sigma_1^2+\sigma_2^2}\left(\sigma_2^4(s-\mu_1)^2 + \sigma_1^4\mu_2^2 + 2\sigma_1^2\mu_2 \cdot \sigma_2^2(s-\mu_1)\right)$$

$$= \frac{\sigma_2^2(s-\mu_1)^2}{\sigma_1^2 \cdot (\sigma_1^2+\sigma_2^2)} + \frac{\sigma_1^2(\mu_2)^2}{\sigma_2^2(\sigma_1^2+\sigma_2^2)} - \frac{2\mu_1(s-\mu_2)}{\sigma_1^2+\sigma_2^2}$$

$$f_S(s) = \frac{1}{2\pi\sigma_1\sigma_2} \exp\left\{ \frac{(s-\mu_1-\mu_2)^2}{2\sigma_1^2+\sigma_2^2} \right\} \underbrace{\int_{-\infty}^{\infty} \exp\left\{ -\frac{1}{2}\left(\frac{\sigma_1^2+\sigma_2^2}{\sigma_1^2\sigma_2^2}\right)\left(t - \frac{\sigma_2^2(s-\mu_1)+\sigma_1^2\mu_2}{\sigma_1^2+\sigma_2^2}\right)^2 \right\} dt}_{\sqrt{\frac{2\sigma_1^2\sigma_2^2\pi}{\sigma_1^2+\sigma_2^2}}}$$

$$\int_{-\infty}^{\infty} \exp\{-ax^2\}\, dx = \sqrt{\frac{\pi}{a}}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{\sigma_1^2+\sigma_2^2}} \exp\left\{ -\frac{1}{2} \cdot \frac{(s-(\mu_1+\mu_2))^2}{\sigma_1^2+\sigma_2^2} \right\}$$

$$= N(\mu_1+\mu_2,\ \sigma_1^2+\sigma_2^2)$$

問6

X, Y , i.i.d. $\sim U(0,1)$

$Z = X+Y$, $W = XY$

$f_X(x) = 1 \quad (0 \leqq x \leqq 1)$

$T = Y$

$J((z,t) \to (x,y)) = 1$

$f_{Z,T}(z,t) = f_X(z-t) f_Y(t)$

$0 < z < 2$

$0 < t < 1$

$0 < z < 1$ のとき $0 < t < z$

$f_Z(z) = \int_0^z 1 \cdot dt = \underline{z}$ //

$1 < z < 2$ のとき $z-1 < t < 1$

$f_Z(z) = \int_{z-1}^1 1 \cdot dt$

$= 1-(z-1) = \underline{2-z}$ //

$W = XY \quad (0 < w < 1)$

$T = Y$

$X = \frac{W}{T}$, $Y = T$ $\quad (0 < \frac{w}{T} < 1)$ $\quad (0 < w < T)$

$$J((w,t) \to (x,y)) = \begin{pmatrix} \frac{1}{T} & -\frac{W}{T^2} \\ 0 & 1 \end{pmatrix}$$

$$= \frac{1}{T}$$

$f_{W,T}(w,t) = f_X(\frac{w}{t}) f_Y(t) \cdot \frac{1}{T}$

$f_W(w) = \int f_X(\frac{w}{t}) f_Y(t) \cdot \frac{1}{t} dt$

$f_W(w) = \int_w^1 \frac{1}{t} dt$

$= [\log t]_w^1 = \underline{-\log w}$ //

問7

$X_1, \cdots, X_n$ は互いに独立

$X_i \sim Po(\lambda_i)$

$\sum_{i=1}^n X_i$ の従う分布

$\varphi_{X_i}(t) = \exp\{(e^{it}-1)\lambda_i\}$

$E[e^{it\sum_{i=1}^n X_i}]$

$= E[e^{it(X_1+X_2+\cdots+X_n)}]$

$= E[e^{itX_1} \cdot e^{itX_2} \cdot \cdots \cdot e^{itX_n}]$

$= E[e^{itX_1}] \cdot \cdots \cdot E[e^{itX_n}]$

$= \varphi_{X_1} \cdot \varphi_{X_2} \cdot \cdots \cdot \varphi_{X_n}$

$= \exp\{\sum_{i=1}^n (e^{it}-1)\lambda_i\}$

$= \exp\{(e^{it}-1)(\lambda_1+\lambda_2+\cdots+\lambda_n)\}$

よって

$\sum_{i=1}^n X_i \sim Po(\lambda_1+\cdots+\lambda_n)$ //

問8

$U_1 \sim U(0,1)$
$U_2 \sim U(0,1)$ } 独立

$r = \sqrt{-2\log U_1}$ 　　 $r^2 = -2\log U_1$

$\theta = 2\pi U_2$ 　　 $-\frac{r^2}{2} = \log U_1$

$U_1 = \exp\{-\frac{r^2}{2}\}$

$$f_{r,\theta}(r,\theta) = f_{U_1}(\exp\{-\tfrac{r^2}{2}\}) \cdot f_{U_2}(\tfrac{\theta}{2\pi}) \cdot |J((r,\theta)\to(u_1,u_2))|$$

$$J((r,\theta)\to(u_1,u_2)) = \begin{pmatrix} -r\exp\{-\frac{r^2}{2}\} & 0 \\ 0 & \frac{1}{2\pi} \end{pmatrix}$$

$$= -\frac{r}{2\pi}\exp\{-\tfrac{r^2}{2}\}$$

$$f_{r,\theta}(r,\theta) = f_{U_1}(\exp\{-\tfrac{r^2}{2}\}) \cdot f_{U_2}(\tfrac{\theta}{2\pi}) \cdot \frac{r}{2\pi}\exp\{-\tfrac{r^2}{2}\}$$

$$= \frac{r}{2\pi}\exp\{-\tfrac{r^2}{2}\}$$

$0 < u_1 < 1$

$0 < u_2 < 1$ 　　 $\frac{y}{x} = \tan\theta$

$\theta = \arctan\frac{y}{x}$

$-\infty < \log u_1 < 0$

$0 < -2\log u_1 < \infty$

$0 < r < \infty$ 　　 $x^2+y^2 = r^2$

$r = \sqrt{x^2+y^2}$

$x = r\cos\theta$
$y = r\sin\theta$

$$f_{X,Y}(x,y) = f_{r,\theta}(\sqrt{x^2+y^2}, \arctan\tfrac{y}{x}) \cdot |J((x,y)\to(r,\theta))|$$

$$J((r,\theta)\to(x,y)) = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}$$

$$= r\cos^2\theta + r\sin^2\theta = r$$

$$f_{X,Y}(x,y) = f_r(\sqrt{x^2+y^2}) \cdot f_\theta(\arctan\tfrac{y}{x}) \cdot \frac{1}{\sqrt{x^2+y^2}}$$

$$f_{X,Y}(x,y) = \frac{1}{\sqrt{x^2+y^2}} \cdot \frac{\sqrt{x^2+y^2}}{2\pi} \cdot \exp\{-\tfrac{x^2+y^2}{2}\}$$

$$= \frac{1}{2\pi}\exp\{-\tfrac{x^2+y^2}{2}\}$$

$$= \frac{1}{\sqrt{2\pi}}\exp\{-\tfrac{x^2}{2}\} \cdot \frac{1}{\sqrt{2\pi}}\exp\{-\tfrac{y^2}{2}\}$$

よって 独立で $N(0,1)$ に従う

問 9

$$\left.\begin{aligned} r^2 &\sim \chi^2_2 \\ \theta &\sim U(0, 2\pi) \end{aligned}\right\} \text{独立}$$

$X = r\cos\theta$, $Y = r\sin\theta$ $\quad \frac{y}{x} = \tan\theta$

$$f_{X,Y}(x,y) = f_{r^2}(x^2+y^2) \cdot f_\theta\left(\arctan\frac{y}{x}\right) \cdot \left|J\left((x,y) \to (r^2,\theta)\right)\right|$$

$$J\left((x,y) \to (r^2,\theta)\right) = \frac{1}{J\left((r^2,\theta) \to (x,y)\right)}$$

$(r^2)^{\frac{1}{2}} \cdot \cos\theta$

$$J\left((r^2,\theta) \to (x,y)\right) = \begin{pmatrix} \frac{\cos\theta}{2r} & -r\sin\theta \\ \frac{\sin\theta}{2r} & r\cos\theta \end{pmatrix}$$

$$= \frac{\cos^2\theta}{2} + \frac{\sin^2\theta}{2} = \frac{1}{2}$$

$$f_{X,Y}(x,y) = 2 \cdot \frac{1}{\Gamma(1)}\left(\frac{1}{2}\right) \cdot \exp\left\{-\frac{x^2+y^2}{2}\right\} \cdot \frac{1}{2\pi}$$

$$= \frac{1}{2\pi}\exp\left\{-\frac{x^2+y^2}{2}\right\}$$

$$= \frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{x^2}{2}\right\} \cdot \frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{y^2}{2}\right\} \quad //$$

問10

$X \sim N(\mu, \sigma^2)$

$Y \sim N(\xi, \sigma^2)$

$U = X + Y$

$V = X - Y$

$X = \dfrac{U+V}{2}$

$Y = \dfrac{U-V}{2}$

$$f_{U,V}(u,v) = f_X\left(\frac{u+v}{2}\right) \cdot f_Y\left(\frac{u-v}{2}\right) \cdot |J|$$

$$J((u,v) \to (x,y)) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

$$= -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}$$

$$f_{U,V}(u,v) = \frac{1}{2} \cdot \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{\left(\frac{u+v}{2} - \mu\right)^2}{2\sigma^2}\right\}$$

$$\cdot \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{\left(\frac{u-v}{2} - \xi\right)^2}{2\sigma^2}\right\}$$

$$= \frac{1}{4\pi\sigma^2} \exp\left\{-\frac{1}{2\sigma^2}\left[\left(\frac{u+v}{2} - \mu\right)^2 + \left(\frac{u-v}{2} - \xi\right)^2\right]\right\}$$

問11

$X \sim N(0, \sigma^2)$

$Y \sim N(0, \sigma^2)$ i.i.d

$Y = \pm\sqrt{w(1-z^2)}$

$w = X^2 + Y^2$

$z = \dfrac{X}{\sqrt{w}}$

$w = wz^2 + Y^2$

$Y^2 = w(1-z^2)$

$$f_{z,w}(z,w) = f_X(\sqrt{w}z)\, f_Y(+\sqrt{w(1-z^2)}) \cdot |J|$$

$$+\, f_X(\sqrt{w}z)\, f_Y(-\sqrt{w(1-z^2)}) \cdot |J|$$

$$J\big((z,w) \to (x,y)\big)$$

$$= \begin{pmatrix} \sqrt{w} & \dfrac{z}{2\sqrt{w}} \\ -wz\,(w(1-z^2))^{-\frac{1}{2}} & \dfrac{\sqrt{1-z^2}}{2\sqrt{w}} \end{pmatrix}$$

$$= \frac{\sqrt{1-z^2}}{2} + wz^2 \cdot \frac{1}{2\sqrt{w(1-z^2)} \cdot \sqrt{w}}$$

$$= \frac{\sqrt{1-z^2}}{2} + \frac{z^2}{2\sqrt{1-z^2}} = \frac{1}{2\sqrt{1-z^2}}$$

$$f_{z,w}(z,w) \overset{2\times}{=} \frac{1}{2\sqrt{1-z^2}} \cdot \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{wz^2}{2\sigma^2}\right\}$$

$$\cdot \frac{1}{\sqrt{2\pi}\sigma} \cdot \exp\left\{-\frac{w(1-z^2)}{2\sigma^2}\right\}$$

$$= \frac{1}{2\cancel{4}\pi\sigma^2} \cdot \frac{1}{\sqrt{1-z^2}} \cdot \exp\left\{-\frac{w}{2\sigma^2}\right\}$$

$-wz^2$

$(w(1-z^2))^{\frac{1}{2}}$

$= \frac{1}{2}(w(1-z^2))^{-\frac{1}{2}} \cdot -2wz$

$= -wz($

$(\sqrt{1-z^2} \cdot w^{\frac{1}{2}})'$

$= \dfrac{\sqrt{1-z^2}}{2\sqrt{w}}$

$-w\cancel{z^2} - w + w\cancel{z^2}$

問12

$X, Y \sim N(0,1)$ i.i.d

$X \cdot \dfrac{1}{X+Y}$

$\underbrace{X = T}$ $\dfrac{X}{X+Y} = S$ $S = \dfrac{T}{T+Y}$

$S(T+Y) = T$

$ST + SY = T$

$SY = T(1-S)$

$\underbrace{Y = \frac{T}{S}(1-S)}$

$f_{S,T}(s,t) = f_X(t)\, f_Y\left(\frac{t}{s} - t\right) \ |J|$

$t \cdot s^{-1}$

$J\left((s,t) \to (x,y)\right)$

$$= \begin{pmatrix} 0 & 1 \\ -\frac{t}{s^2} & \end{pmatrix} = -\frac{t}{s^2}$$

$$f_{S,T}(s,t) = \frac{1}{\sqrt{2\pi}} \exp\left\{\frac{t^2}{2}\right\} \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{\left(\frac{t}{s}-t\right)^2}{2}\right\} \cdot \frac{t}{s^2}$$

$$= \frac{1}{2\pi} \cdot \frac{t}{s^2} \exp\left\{-\frac{t^2 + \left(\frac{t}{s}-t\right)^2}{2}\right\}$$

$$f_S(s) = \int_{-\infty}^{\infty} \frac{1}{2\pi} \cdot \frac{t}{s^2} \exp\left\{-\frac{t^2+\left(\frac{t}{s}-t\right)^2}{2}\right\} dt$$

$$= \left[\frac{-1}{2\pi} \cdot \frac{1}{s^2} \cdot \frac{1}{\left(1+\frac{1}{s^2}-\frac{2}{s}+1\right)} \exp\{\ \}\right]_{-\infty}^{\infty}$$

$= 0$

$$-\frac{t^2 + \left(\frac{t}{s}-t\right)^2}{2}$$

$$= -\frac{t^2 + \frac{t^2}{s^2} - \frac{2t^2}{s} + t^2}{2}$$

$$= -\frac{t^2}{2}\left(1 + \frac{1}{s^2} - \frac{2}{s} + 1\right)$$

問12

$X, Y \sim N(0,1)$ i.i.d

①
$$
\begin{aligned}
F_Z(z) &= P(Z \leq z) \\
&= P\left(\frac{X}{|Y|} \leq z\right) \\
&= P(X \leq |Y| z) \\
&= P(X \leq Yz, Y \geq 0) \\
&\quad + P(X \leq -Yz, Y < 0) \\
&= \int_0^\infty \left(\int_{-\infty}^{Yz} f_X(x)\,dx \right) f_Y(y)\,dy \\
&\quad + \int_{-\infty}^0 \left(\int_{-\infty}^{-Yz} f_X(x)\,dx \right) f_Y(y)\,dy
\end{aligned}
$$

$$
\begin{aligned}
f_Z(z) &= \int_0^\infty y\, f_X(yz) f_Y(y) \\
&\quad - \int_{-\infty}^0 y\, f_X(-yz) f_Y(y)\,dy \\
&= 2\int_0^\infty y\, f_X(yz) f_Y(y)\,dy \\
&= 2\int_0^\infty y \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{y^2 z^2}{2}\right\} \\
&\qquad \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{y^2}{2}\right\} dy \\
&= 2\int_0^\infty \frac{1}{2\pi}\, y \exp\left\{-\frac{y^2(z^2+1)}{2}\right\} dy \\
&= 2\left[\frac{-1}{z^2+1} \frac{1}{2\pi} \exp\left\{-\frac{y^2(z^2+1)}{2}\right\} \right]_0^\infty \\
&= \frac{1}{\pi} \cdot \frac{1}{z^2+1} \quad //
\end{aligned}
$$

②
$$
W = \frac{X}{X+Y} = \frac{\frac{X}{Y}}{\frac{X}{Y}+1} = \frac{Z}{Z+1}
$$

$$
\begin{aligned}
F_W(w) &= P(W \leq w) \\
&= P\left(\frac{Z}{Z+1} \leq w\right) \\
&= P\left(Z \leq \frac{w}{1-w}\right)
\end{aligned}
$$

$$
\begin{aligned}
f_W(w) &= f_Z\left(\frac{w}{1-w}\right) \cdot \left| \frac{d}{dw}\left(\frac{w}{1-w}\right) \right| \\
&= \frac{1}{(1-w)^2} \cdot \frac{1}{\pi} \cdot \frac{1}{\left(\frac{w}{1-w}\right)^2+1} \\
&= \frac{1}{\pi} \cdot \frac{1}{w^2+(1-w)^2}.
\end{aligned}
$$

問13

$$\lambda e^{-\lambda x},\ x>0$$

$$\left.\begin{array}{l} X \sim Ex(\lambda) \\ Y \sim Ex(\mu) \end{array}\right\}\ 独立$$

$$Z=\min\{X,Y\}$$

$Z=X$ のとき $w=1$

$Z=Y$ のとき $w=0$

同時分布　を求める。

$w=1$ のとき

$$F_{Z,w}(z,w) = P(Z<z,\ w=1)$$
$$= P(Z<z, Z=X, w=1)$$
$$= P(X<z,\ X<Y)$$
$$= P(X<z)\cdot P(X\leq Y)$$
$$= \int_0^z f_X(x)\left(\int_0 f_Y(y)dy\right)dx$$
$$= \int_0^z f_X(x)\left(\int_x^\infty \mu e^{-\mu y}dy\right)dx$$
$$= \int_0^z f_X(x)\cdot\left[-e^{-\mu y}\right]_x^\infty dx$$
$$= \int_0^z \lambda e^{-\lambda x}\cdot(e^{-\mu x})dx$$
$$= \int_0^z \lambda e^{-(\lambda+\mu)x}dx$$
$$= \left[\frac{-\lambda}{\lambda+\mu}e^{-(\lambda+\mu)x}\right]_0^z$$
$$= \frac{-\lambda}{\lambda+\mu}\left(e^{-(\lambda+\mu)z}-1\right)$$

$w=0$ のとき

$$F_{Z,w}(z,w) = P(Z\leq z,\ w=0)$$
$$= P(Z\leq z, Z=Y, w=0)$$
$$= P(Y\leq z,\ X>Y)$$
$$= \int_0^z\left(\int_y^\infty f_X(x)dx\right)f_Y(y)dy$$
$$= \int_0^z\left(\int_y^\infty \lambda e^{-\lambda x}dx\right)f_Y(y)dy$$
$$= \int_0^z\left[-e^{-\lambda x}\right]_y^\infty f_Y(y)dy$$
$$= \int_0^z e^{-\lambda y}\cdot\mu\cdot e^{-\mu y}dy$$
$$= \int_0^z \mu\cdot e^{-(\lambda+\mu)y}dy$$
$$= \left[\frac{-\mu}{\lambda+\mu}e^{-(\lambda+\mu)y}\right]_0^z$$
$$= \frac{-\mu}{\lambda+\mu}\left(e^{-(\lambda+\mu)z}-1\right)$$

問14

$$\left.\begin{aligned} X &\sim \chi^2_{m_1} \\ Y &\sim \chi^2_{m_2} \end{aligned}\right\} \text{独立}$$

(1)

$$Z = X+Y \qquad X = WZ$$

$$W = \frac{X}{X+Y} \qquad Y = Z(1-W)$$

$$f_{Z,W}(z,w) = f_X(wz)\, f_Y(z(1-w))\, |J|$$

$$J((z,w) \to (x,y))$$

$$= \begin{pmatrix} w & z \\ 1-w & -z \end{pmatrix} = -wz - z(1-w)$$

$$= -z$$

$$f_{Z,W}(z,w) = \frac{1}{\Gamma(\frac{m_1}{2})}\left(\frac{1}{2}\right)^{\frac{m_1}{2}} \cdot (wz)^{\frac{m_1}{2}-1}$$

$$\exp\left\{-\frac{wz}{2}\right\} \cdot \frac{1}{\Gamma(\frac{m_2}{2})}\left(\frac{1}{2}\right)^{\frac{m_2}{2}} \cdot (z(1-w))^{\frac{m_2}{2}-1}$$

$$\cdot \exp\left\{-\frac{z(1-w)}{2}\right\} \cdot z$$

$$= \frac{1}{\Gamma(\frac{m_1}{2})\cdot\Gamma(\frac{m_2}{2})} \cdot \left(\frac{1}{2}\right)^{\frac{m_1+m_2}{2}} \cdot z^{\frac{m_1+m_2}{2}-1}$$

$$\cdot \exp\left\{-\frac{z}{2}\right\} \cdot w^{\frac{m_1}{2}-1} \cdot (1-w)^{\frac{m_2}{2}-1}$$

$$= \frac{1}{\Gamma(\frac{m_1+m_2}{2})} \cdot \left(\frac{1}{2}\right)^{\frac{m_1+m_2}{2}} \cdot z^{\frac{m_1+m_2}{2}-1}$$

$$\cdot \exp\left\{-\frac{z}{2}\right\} \cdot \frac{1}{\frac{\Gamma(\frac{m_1}{2})\cdot\Gamma(\frac{m_2}{2})}{\Gamma(\frac{m_1+m_2}{2})}}$$

$$\cdot w^{\frac{m_1}{2}-1}(1-w)^{\frac{m_2}{2}-1}$$

$$= \chi^2_{m_1+m_2} \cdot \mathrm{Beta}\left(\frac{m_1}{2}, \frac{m_2}{2}\right)$$

//

(2)

$$Z \sim \chi^2_{m_1+m_2}$$

$$W \sim \mathrm{Beta}\left(\frac{m_1}{2}, \frac{m_2}{2}\right)$$

//

問15

$\lambda e^{-\lambda x}$ $(\lambda > 0)$

$x > 0$

$X_1, X_2, X_3 \sim Ex(\lambda)$ i.i.d

$Z_1 = \frac{X_1}{X_1+X_2}$, $Z_2 = \frac{X_1+X_2}{X_1+X_2+X_3}$

$Z_3 = X_1+X_2+X_3$

$X_1 + X_2 = Z_2 Z_3$

$X_1 = \underline{Z_1 Z_2 Z_3}$

$X_2 = \underline{Z_2 Z_3 (1-Z_1)}$

$X_3 = Z_3 - Z_2 Z_3$

$= \underline{Z_3(1-Z_2)}$

$f_{Z_1,Z_2,Z_3}(z_1, z_2, z_3)$

$= f_{X_1}(z_1 z_2 z_3) \cdot f_{X_2}(z_2 z_3 (1-z_1))$

$\cdot f_{X_3}(z_3(1-z_2)) \cdot |J|$

$J((z_1, z_2, z_3) \to (x_1, x_2, x_3))$

$$= \begin{pmatrix} Z_2 Z_3 & Z_1 Z_3 & Z_1 Z_2 \\ -Z_2 Z_3 & Z_3(1-Z_1) & Z_2(1-Z_1) \\ 0 & -Z_3 & 1-Z_2 \end{pmatrix}$$

$= Z_2 Z_3^2 (1-Z_1)(1-Z_2) + Z_1 Z_2^2 Z_3^2$

$+ Z_1 Z_2 Z_3^2 (1-Z_2) + Z_2^2 Z_3^2 (1-Z_1)$

$= Z_2 Z_3^2 (1 - Z_1 - Z_2 + Z_1 Z_2 + Z_1 Z_2$

$+ Z_1 - Z_1 Z_2 + Z_2 - Z_1 Z_2)$

$= Z_2 Z_3^2 (1) = Z_2 Z_3^2$

$f_{Z_1 Z_2 Z_3}(z_1, z_2, z_3)$

$= \lambda e^{-\lambda(z_1 z_2 z_3)} \cdot \lambda e^{-\lambda(z_2 z_3 (1-z_1))}$

$\cdot \lambda e^{-\lambda(z_3 - z_2 z_3)} \cdot z_2 \cdot z_3^2$

$= \lambda^3 z_2 z_3^2 \exp\{-\lambda[z_1 z_2 z_3 + z_2 z_3 - z_1 z_2 z_3$

$+ z_3 - z_2 z_3]\}$

$= \lambda^3 z_2 z_3^2 \exp\{-\lambda z_3\}$ $(z_1 = 1)$

より積の形で表されるから独立

(2)

$f_{z_1, z_2}(z_1, z_2) = \int_0^\infty \lambda^3 z_2 z_3^2 \exp\{-\lambda z_3\} dz_3$

$= \left[-\lambda^2 z_2 \cdot z_3^2 \exp\{-\lambda z_3\}\right]_0^\infty + \int_0^\infty 2\lambda^2 z_2 z_3 \exp\{-\lambda z_3\} dz_3$

$= \left[-2\lambda z_2 z_3 \exp\{-\lambda z_3\}\right]_0^\infty + \int_0^\infty 2\lambda z_2 \exp\{-\lambda z_3\} dz_3$

$= \left[-2z_2 \exp\{-\lambda z_3\}\right]_0^\infty = 2 z_2$

$Z_1 = 1$ より一様分布

$Z_2 \sim 2z_2$

(3) $Z_3 \sim \frac{1}{2}\lambda^3 z_3^2 \exp\{-\lambda z_3\}$

問16

(1)

$$Cov(X,Y) = E[Cov(X,Y|Z)] + Cov(E[X|Z], E[Y|Z])$$ を示せ

$$Cov(X,Y) = E[XY] - E[X]E[Y]$$

$$= E[XY] - E[E[X|Z]\cdot E[Y|Z]] + E[E[X|Z]\cdot E[Y|Z]] - E[X]E[Y]$$

$$= E[E[XY|Z]] - E[E[X|Z]\cdot E[Y|Z]] + E[E[X|Z]\cdot E[Y|Z]] - E[E[X|Z]]E[E[Y|Z]]$$

$$= E[E[XY|Z] - E[X|Z]E[Y|Z]] + E[E[X|Z]\cdot E[Y|Z]] - E[E[X|Z]]\cdot E[E[Y|Z]]$$

$$= E[Cov(X,Y|Z)] + Cov(E[X|Z], E[Y|Z])$$

$$Cov(E[X|Z], E[Y|Z])$$

$$= E[E[X|Z]\cdot E[Y|Z]] - E[E[X|Z]]E[E[Y|Z]]$$

$$= E[E[X|Z]\cdot E[Y|Z]] - E[X]\cdot E[Y]$$

$$E[Cov(X,Y|Z)]$$

$$= E[XY] - E[E[X|Z]\cdot E[Y|Z]]$$

$$= E[E[XY|Z] - E[X|Z]E[Y|Z]]$$

$$= E[XY] - E[E[X|Z]\cdot E[Y|Z]]$$

(2)

$Z \sim N(0,1)$

$$\left.\begin{array}{l} X \mid Z \sim N(z,1) \\ Y \mid Z \sim N(z,1) \end{array}\right\} \text{独立}$$

$\mathrm{Cov}(X,Y)$

$$= \underbrace{E[\mathrm{Cov}(X,Y \mid Z)]}_{0} + \mathrm{Cov}(E[X \mid Z], E[Y \mid Z])$$

$$= \mathrm{Cov}(Z, Z)$$

$$= E[Z^2] - E[Z]$$

$$= 1 - 0 = \underline{1} \text{ (ア)}$$

$$\mathrm{Var}(Z) = E[Z^2] - E[Z]^2$$

$$= \sim - 0 = 1$$

問17

$$Y \mid X \sim N(\alpha + \beta X, 1)$$

$$X \sim N(\mu, 1)$$

(1)

$$\mathrm{Var}(Y) = E[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(E[Y \mid X])$$

$$= 1 + \mathrm{Var}(\alpha + \beta X)$$

$$= 1 + \beta^2 \mathrm{Var}(X)$$

$$= \underline{1 + \beta^2}$$

(2) XとYの相関係数

$$\rho = \frac{E[XY] - E[X]E[Y]}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}$$

$$= \frac{E[XY] - \mu \cdot E[E[Y \mid X]]}{\sqrt{1+\beta^2}}$$

$$= \frac{E[XY] - \mu \cdot E[\alpha + \beta X]}{\sqrt{1+\beta^2}}$$

$$= \frac{E[XY] - \mu \cdot (\alpha + \beta\mu)}{\sqrt{1+\beta^2}}$$

$$E[XY]$$

xが与えられたときのXは定数

$$= E[E[XY \mid X]]$$

$$= E[X\,E[Y \mid X]]$$

$$= E[X \cdot (\alpha + \beta X)]$$

$$= E[\beta X^2 + \alpha X]$$

$$= \beta E[X^2] + \alpha E[X] = \beta(1 + \mu^2) + \alpha\mu$$

$$E[X^2] - E[X]^2 = 1$$

$$E[X^2] = 1 + \mu^2$$

$$\rho = \frac{\beta + \beta\mu^2 + \alpha\mu - \alpha\mu - \beta\mu^2}{\sqrt{1+\beta^2}}$$

$$= \underline{\frac{\beta}{\sqrt{1+\beta^2}}}$$

(3)

$f_{X|Y}$ を求めよ

$$f_{X|Y} = \frac{f_{X,Y}}{f_Y} = \frac{f_{Y|X} \cdot f_X}{f_Y}$$

$$= \frac{\frac{1}{\sqrt{2\pi}} \cdot \exp\left\{-\frac{(y-(\alpha+\beta x))^2}{2}\right\} \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{(x-\mu)^2}{2}\right\}}{\int_{-\infty}^{\infty} \frac{1}{2\pi} \exp\left\{-\frac{(y-(\alpha+\beta x))^2+(x-\mu)^2}{2}\right\} dx}$$

$$y^2 - 2y(\alpha+\beta x) + \alpha^2 + 2\alpha\beta x$$
$$+ \beta^2 x^2 + x^2 - 2\mu x + \mu^2$$

$$= (\beta^2+1)x^2 + 2(\alpha\beta - y\beta - \mu)x$$
$$+ y^2 - 2y\alpha + \alpha^2 + \mu^2$$

$$= (\beta^2+1)\left(x + \frac{\alpha\beta - y\beta - \mu}{(\beta^2+1)}\right)^2$$
$$- \frac{(\alpha\beta - y\beta - \mu)^2}{(\beta^2+1)} + y^2 - 2y\alpha + \alpha^2 + \mu^2$$

$$f_Y = \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp\left\{-\frac{\beta^2+1}{2}(x+\quad)^2\right\} dx$$
$$\times \exp\{\qquad\}$$

$$= \frac{1}{2\pi} \times \sqrt{\pi} \times \sqrt{\frac{2}{\beta^2+1}}$$
$$\times \exp\left\{-\frac{(\alpha\beta - y\beta - \mu)^2}{(\beta^2+1)} + y^2 - 2y\alpha + \alpha^2 + \mu^2\right\}$$

$$f_{X|Y} = \frac{\sqrt{\beta^2+1}}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}(\beta^2+1)\left(x + \frac{\alpha\beta - y\beta - \mu}{\beta^2+1}\right)^2\right\}$$

$$= \frac{1}{\sqrt{2\pi} \cdot \frac{1}{\sqrt{\beta^2+1}}} \exp\left\{-\frac{\left(x - \frac{-\alpha\beta + y\beta + \mu}{\beta^2+1}\right)^2}{2 \cdot \frac{1}{\beta^2+1}}\right\}$$

$$= \mathcal{N}\left(\frac{\mu + \beta(y-\alpha)}{\beta^2+1}, \frac{1}{\beta^2+1}\right)$$

//

問18

$x>0$, $g(x)$ が

$g(x) \geq 0$ と $\int_0^\infty g(x)\,dx = 1$ を満たす.

(1)

$$A = \{(x,y) \mid x>0, y>0\}$$

$$f(x,y) = C \times \frac{g(\sqrt{x^2+y^2})}{\sqrt{x^2+y^2}}$$

$$\int_0^\infty \int_0^\infty \frac{g\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}}\,dx\,dy$$

$$x = r\cos\theta \quad y = r\sin\theta \quad \left(r>0,\ 0<\theta<\frac{\pi}{2} \right)$$

$$J((r,\theta)\to(x,y)) = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}$$

$$= r\cos^2\theta + r\sin^2\theta = r$$

$$f_{r,\theta}(r,\theta) = f_{x,y}(r\cos\theta, r\sin\theta)\cdot r$$

$$= C\cdot\frac{g(r)}{r}\cdot r$$

$$= C\cdot g(r)$$

$$\int_0^\infty \int_0^{\frac{\pi}{2}} C\cdot g(r)\cdot d\theta\cdot dr$$

$$= \int_0^\infty \frac{\pi}{2}\cdot C\cdot g(r)\cdot dr$$

$$= \frac{\pi}{2}C = 1$$

$$\therefore C = \frac{2}{\pi}$$

(2)

$$W = \frac{X^2}{X^2+Y^2} \qquad X^2 = WZ$$

$$X = \sqrt{WZ}$$

$$Z = X^2+Y^2$$

$$Y^2 = Z - WZ$$

$$= Z(1-W)$$

$$Y = \sqrt{Z(1-W)}$$

$$\begin{pmatrix} W>0 \\ Z>0 \\ 1-W>0 \therefore W<1 \end{pmatrix}$$

$$\hookrightarrow 0<W<1$$
$$0<Z$$

$$f_{Z,W}(z,w) = f_{X,Y}(\sqrt{wz}, \sqrt{z(1-w)})\cdot |J|$$

$$w^{\frac{1}{2}}\cdot z^{\frac{1}{2}} \qquad \sqrt{z}\cdot\sqrt{1-w}$$

$$J((z,w)\to(x,y))$$

$$= \begin{pmatrix} \frac{1}{2}w^{\frac{1}{2}}\cdot z^{-\frac{1}{2}} & \frac{1}{2}w^{-\frac{1}{2}}z^{\frac{1}{2}} \\ \frac{1}{2}\sqrt{1-w}\cdot z^{-\frac{1}{2}} & \frac{-1}{2}\cdot z^{\frac{1}{2}}\cdot(1-w)^{-\frac{1}{2}} \end{pmatrix}$$

$$= -\frac{1}{4}\frac{\sqrt{w}}{\sqrt{1-w}} - \frac{1}{4}\frac{\sqrt{1-w}}{\sqrt{w}}$$

$$= -\frac{1}{4}\left(\frac{1}{\sqrt{w(1-w)}}\right)$$

$$f_{Z,W}(z,w) = \frac{1}{4}\cdot\frac{1}{\sqrt{w(1-w)}}\cdot\frac{2}{\pi}\cdot\frac{g(\sqrt{z})}{\sqrt{wz+z(1-w)}}$$

$$= \frac{1}{2\pi}\cdot\frac{1}{\sqrt{w(1-w)}}\cdot\frac{g(\sqrt{z})}{\sqrt{z}}$$

積の形で表せれるから独立

$$f_W(w) = \int_0^\infty \frac{1}{2\pi}\cdot\frac{1}{\sqrt{w(1-w)}}\cdot\frac{g(\sqrt{z})}{\sqrt{z}}\cdot dz$$

$$= \frac{1}{2\pi\sqrt{w(1-w)}}\int_0^\infty \frac{g(\sqrt{z})}{\sqrt{z}}\cdot dz$$

$$\sqrt{z} = t \qquad \frac{dt}{dz} = \frac{1}{2}\cdot\frac{1}{\sqrt{z}}$$

$$2dt = \frac{1}{\sqrt{z}}\cdot dz$$

$$= \frac{2}{2\pi\sqrt{w(1-w)}}\int_0^\infty g(t)\cdot dt$$

$$= \frac{1}{\pi\sqrt{w(1-w)}}$$

$$f_Z(z) = \frac{g(\sqrt{z})}{2\sqrt{z}}$$

問19

$X|Y \sim Bin(n, Y)$

$Y \sim Beta(\alpha, \beta)$

$$f_X(x) = \binom{n}{x} \cdot \frac{B(x+\alpha, n-x+\beta)}{B(\alpha, \beta)}$$

$x = 0, \cdots, n$

平均

$$E[X] = E[E[X|Y]]$$

$$= E^Y[nY]$$

$$= n E[Y]$$

$$= n \cdot \frac{\alpha}{\alpha+\beta}$$

分散

$$Var(X) = E[Var(X|Y)] + Var(E[X|Y])$$

$$= E[nY(1-Y)] + Var(nY)$$

$$= nE[Y] - nE[Y^2] + n^2 Var(Y)$$

$$Var(Y) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$$E[Y^2] - E[Y]^2 = Var(Y)$$

$$E[Y^2] = Var(Y) + E[Y]^2$$

$$= \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} + \frac{\alpha^2}{(\alpha+\beta)^2}$$

$$= \frac{\alpha\beta + \alpha^2(\alpha+\beta+1)}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$Var(X)$

$$= n \cdot \frac{\alpha}{\alpha+\beta} - n \cdot \frac{\alpha\beta + \alpha^2(\alpha+\beta+1)}{(\alpha+\beta)^2(\alpha+\beta+1)} + n^2 \cdot \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$$= \frac{n\alpha(\alpha+\beta)(\alpha+\beta+1) - n\alpha\beta - n\alpha^2(\alpha+\beta+1) + n^2\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$$= \frac{n\alpha\beta(\alpha+\beta) + n^2\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$$= \frac{n\alpha\beta(\alpha+\beta+n)}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

問20

$X|Y \sim Po(Y)$

$Y \sim Ga(\alpha, \beta)$

$$f_X(x) = \frac{\Gamma(x+\alpha)}{\Gamma(\alpha)x!} \frac{\beta^x}{(1+\beta)^{x+\alpha}}$$

平均

$$E[X] = E[E[X|Y]]$$

$$= E[Y]$$

$$= \alpha\beta$$ //

分散

$$Var(X) = E[Var(X|Y)] + Var(E[X|Y])$$

$$= E[Y] + Var(Y)$$

$$= \alpha\beta + \alpha\beta^2$$

$$= \alpha\beta(1+\beta)$$ //

問21

$$f(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}(x^2-2\rho xy+y^2)\right\}$$

(1) $U = X,\ V = Y-\rho X$

$X = U,\quad Y = V+\rho U$

$$J((u,v)\to(x,y)) = \begin{pmatrix} 1 & 0 \\ \rho & 1 \end{pmatrix} = 1$$

$$f_{U,V}(u,v) = f_{X,Y}(u, v+\rho u)\,|J|$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left(u^2-2\rho u(v+\rho u) + (v+\rho u)^2\right)\right\}$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left((1-\rho^2)u^2+v^2\right)\right\}$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{u^2}{2}\right\} \cdot \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \exp\left\{-\frac{v^2}{2(1-\rho^2)}\right\}$$

$$= N(0,1)\cdot N(0,1-\rho^2)$$

よって　UとVは独立で

$U \sim N(0,1)$

$V \sim N(0,1-\rho^2)$

(2)

$$\mathrm{Corr}(X,Y) = \frac{E[XY]-E[X]E[Y]}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}}$$

(1)より　$E[U]=0 \qquad E[V]=0$

$\mathrm{Var}(U)=1 \qquad \mathrm{Var}(V) = 1-\rho^2$

よって

$E[U]=E[X]=0 \qquad \mathrm{Var}(U)=\mathrm{Var}(X)=1$

$E[Y-\rho X]$

$= E[Y]-\rho E[X]$

$\therefore E[Y]=0$

$\mathrm{Var}(Y) = \mathrm{Var}(V+\rho U)$

$= \mathrm{Var}(V)+\rho^2\mathrm{Var}(U)$

$= 1-\rho^2+\rho^2 = 1$

$$E[XY] = E[U(V+\rho U)]$$
$$= E[UV]+\rho E[U^2]$$
$$= E[U]\cdot E[V] + \rho\left(\mathrm{Var}(U)+E[U]^2\right)$$
$$= \rho$$

よって

$$\mathrm{Corr}(X,Y) = \frac{\rho}{\sqrt{1}\cdot\sqrt{1}} = \rho$$

$$\mathrm{Corr}(X^2,Y^2) = \frac{E[X^2Y^2]-E[X^2]E[Y^2]}{\mathrm{Var}(X^2)\,\mathrm{Var}(Y^2)}$$

$$E[X^2] = E[U^2] = \mathrm{Var}(U)+E[U]^2 = 1$$

$$\mathrm{Var}(X^2) = \mathrm{Var}(U^2) = E[U^4]-E[U^2]^2$$

$(U^2 \sim \chi^2_1)$ $\qquad = 3-1 = 2$

$$E[Y^2] = E[(V+\rho U)^2] = E[V^2+\rho UV+\rho^2U^2]$$
$$= E[V^2]+\rho E[U]E[V] + \rho^2 E[U^2]$$
$$= 1-\rho^2+\rho^2 = 1$$

$\mathrm{Var}(V) = E[V^2]$

$E[X^2] = 1$, $\quad var(X^2) = 2$

$E[Y^2] = 1$

$E[U^4] = 3$

$$Var(Y^2) = Var((V+\rho U)^2)$$
$$= Var(V^2 + 2\rho V U + \rho^2 U^2)$$
$$= Var(V^2) + 4\rho^2 Var(VU) + \rho^4 Var(U^2)$$
$$= 3(1-\rho^2)^2 - (1-\rho^2)^2$$
$$+ 4\rho^2 \left(E[V^2]E[U^2] - E[V]^2 E[U]^2 \right)$$
$$+ E[U^4] - E[U^2]^2$$
$$= 2(1-\rho^2)^2 + 4\rho^2 ((1-\rho^2))$$
$$+ \rho^4 (3-1)$$
$$= 2 - 4\rho^2 + 2\rho^4 + 4\rho^2 - 4\rho^4$$
$$+ 2\rho^4$$
$$= \underline{2}$$

$$E[X^2 Y^2]$$
$$= E[U^2(V^2 + 2\rho U V + \rho^2 U^2)]$$
$$= E[U^2 V^2] + 2\rho E[U^3 V]$$
$$+ \rho^2 E[U^4]$$
$$= 1-\rho^2 + 3\rho^2 = 2\rho^2 + 1$$

$$E[V^2] = Var(V) + E[V]^2$$
$$= 1 - \rho^2$$

$$E[U^2] = 1$$

$$\int_{-\infty}^{\infty} v^4 \cdot \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \exp\left\{-\frac{v^2}{2(1-\rho^2)}\right\} dv$$
$$= \left[v^3 \cdot \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{1-\rho^2}} (-(1-\rho^2)) \cdot \exp\left\{-\frac{v^2}{2(1-\rho^2)}\right\} \right]_{-\infty}^{\infty}$$
$$= \int_{-\infty}^{\infty} 3v^2 \cdot \frac{(1-\rho^2)}{\sqrt{2\pi}\sqrt{1-\rho^2}} \exp\left\{-\frac{v^2}{2(1-\rho^2)}\right\} dv$$
$$= \left[-3v \cdot \frac{(1-\rho^2)^2}{\sqrt{2\pi}\sqrt{1-\rho^2}} \cdot \exp\left\{-\frac{v^2}{2(1-\rho^2)}\right\} \right]_{-\infty}^{\infty}$$
$$+ \int_{-\infty}^{\infty} 3 \frac{(1-\rho^2)^2}{\sqrt{2\pi}\sqrt{1-\rho^2}} \exp\left\{-\frac{v^2}{2(1-\rho^2)}\right\} dv$$
$$= 3(1-\rho^2)^2 \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \exp\{- \quad\} dv$$
$$= 3(1-\rho^2)^2$$

$$Corr(X^2, Y^2)$$
$$= \frac{(2\rho^2+1) - 1}{\sqrt{2} \cdot \sqrt{2}}$$
$$= \frac{2\rho^2}{2} = \underline{\rho^2} \; //$$

(3)

$$\frac{X^2-2\rho XY+Y^2}{1-\rho^2}$$

$$=\frac{U^2-2\rho U(V+\rho U)+(V+\rho U)^2}{1-\rho^2}$$

$$=\frac{U^2-\cancel{2\rho UV}-2\rho^2U^2+V^2+\cancel{2\rho VU}+\rho^2U^2}{1-\rho^2}$$

$$=\frac{(1-\rho^2)U^2+V^2}{1-\rho^2}=U^2+\frac{V^2}{1-\rho^2}$$

$$=U^2+\left(\frac{V}{\sqrt{1-\rho^2}}\right)^2$$

$V\sim N(0,1)$

$\frac{V}{\sqrt{1-\rho^2}}\sim N(0,1)$ より

$U^2, \frac{V}{\sqrt{1-\rho^2}}\sim \chi^2_1$ i.i.d

よって

分布の再生性より

$$U^2+\left(\frac{V}{\sqrt{1-\rho^2}}\right)^2\sim\chi^2_2$$

//

$V=\sqrt{1-\rho^2}\,T$

$T=\frac{V}{\sqrt{1-\rho^2}}$

$V\sim N(0,1-\rho^2)$

$$\frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}}\exp\left\{-\frac{v^2}{2(1-\rho^2)}\right\}$$

問22

$$f(x,y,z) = \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)} x^{a-1} y^{b-1} z^{c-1}$$

$0<x,y,z<1,\ x+y+z=1$

$a>0, b>0, c>0$

(1) $f(x,y,z)$ が確率密度関数を確かめる

$$= \int_0^1 \int_0^{1-x} x^{a-1} y^{b-1} (1-x-y)^{c-1} dx\, dy$$

$$= \int_0^1 \int_0^{1-x} x^{a-1} y^{b-1} \left((1-x)\cdot\left(1-\frac{y}{1-x}\right)\right)^{c-1} dx dy$$

$$= \int_0^1 x^{a-1}(1-x)^{c}\left(\int_0^{1-x} y^{b-1}\left(1-\frac{y}{1-x}\right)^{c-1} dy\right) dx$$

$$= \int_0^1 x^{a-1}(1-x)^{b+c-1} dx \int_0^1 t^{b-1}(1-t)^{c-1} dt$$

$(1-x)t = y$

$$= B(a,b+c)\ B(b,c)$$

$$= \frac{\Gamma(a)\Gamma(b+c)}{\Gamma(a+b+c)} \times \frac{\Gamma(b)\Gamma(c)}{\Gamma(b+c)}$$

$$= \frac{\Gamma(a)\Gamma(b)\Gamma(c)}{\Gamma(a+b+c)}$$

よって

1

(2)

$$f_X(x) = \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)}\int_0^{1-x} x^{a-1} y^{b-1} (1-x-y)^{c-1} dy$$

$$= \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)}\int_0^{1-x} x^{a-1} y^{b-1} \left((1-x)\left(1-\frac{y}{1-x}\right)\right)^{c-1} dy$$

$$= \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)} x^{a-1}(1-x)^{c-1} \int_0^{1-x} y^{b-1}\left(1-\frac{y}{1-x}\right)^{c-1} dy$$

ここで

$$\int_0^{1-x} y^{b-1}\left(1-\frac{y}{1-x}\right)^{c-1} dy \qquad t = \frac{y}{1-x} とおく$$

$$\frac{dt}{dy} = \frac{1}{1-x} \qquad (1-x)dt = dy$$

$$\int_0^{1-x} (1-x)\cdot(1-x)^{b-1}\cdot t^{b-1}(1-t)^{c-1} dt$$

よって

$$f_X(x) = \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)} x^{a-1}(1-x)^{b+c-1} \int_0^1 t^{b-1}(1-t)^{c-1} dt$$

$$= \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)} x^{a-1}(1-x)^{b+c-1} \cdot \frac{\Gamma(b)\Gamma(c)}{\Gamma(b+c)}$$

$$= \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b+c)} x^{a-1}(1-x)^{b+c-1}$$

$$= \frac{1}{B(a,b+c)} x^{a-1}(1-x)^{b+c-1}$$

$$= \mathrm{Beta}(a,b+c)$$

$$f_Y(y) = \frac{\Gamma(a+b+c)}{\Gamma(b)\Gamma(a+c)} y^{b-1}(1-y)^{a+c-1}$$

$$= \mathrm{Beta}(b, a+c-1)$$

(3)

$$f_{Y|X} = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

$$= \frac{\frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)} x^{a-1} y^{b-1} (1-x-y)^{c-1}}{\frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b+c)} x^{a-1} (1-x)^{b+c-1}}$$

$$= \frac{\Gamma(b+c)}{\Gamma(b)\Gamma(c)} \left(\frac{y}{1-x}\right)^{b-1} \cdot \frac{1}{(1-x)^c} \cdot \left((1-x)\left(1-\frac{y}{1-x}\right)\right)^{c-1}$$

$$= \frac{\Gamma(b+c)}{\Gamma(b)\Gamma(c)} \left(\frac{y}{1-x}\right)^{b-1} \frac{1}{1-x} \cdot \left(1-\frac{y}{1-x}\right)^{c-1}$$

//

(4)

$$E[XY] - E[X]E[Y]$$

$X \sim \mathrm{Beta}(a, b+c)$

$Y \sim \mathrm{Beta}(b, a+c)$

$$E[X] = \frac{a}{a+b+c}, \quad E[Y] = \frac{b}{a+b+c}$$

$$Var(X) = \frac{a(b+c)}{(a+b+c)^2(a+b+c+1)}$$

$$Var(Y) = \frac{b(a+c)}{(a+b+c)^2(a+b+c+1)}$$

$$E[XY] = \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)} \int_0^1 \int_0^{1-x} x^a y^b (1-x-y)^{c-1} dy\, dx$$

$$= \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)} \int_0^1 x^a \int_0^{1-x} y^b (1-x-y)^{c-1} dy\, dx$$

ここで

$(1-x)\left(1-\frac{y}{1-x}\right)$

$\frac{y}{1-x} = t$

$\frac{dt}{dy} = \frac{1}{1-x}$

$$\int_0^{1-x} y^b (1-x-y)^{c-1} dy$$

$$= \int_0^1 (1-x)^{b+1} t^b (1-x)^{c-1} \cdot (1-t)^{c-1} dt$$

$$= (1-x)^{b+c} \int_0^1 t^{(b+1)-1} (1-t)^{c-1} dt$$

$$= (1-x)^{b+c} \frac{\Gamma(b+1)\Gamma(c)}{\Gamma(b+c+1)}$$

よって

$E[XY]$

$$= \frac{\Gamma(a+b+c)\Gamma(b+1)}{\Gamma(a)\Gamma(b)\Gamma(b+c+1)} \int_0^1 x^{(a+1)-1} (1-x)^{(b+c+1)-1} dx$$

$$= \frac{\Gamma(a+b+c)\Gamma(b+1)}{\Gamma(a)\Gamma(b)\Gamma(b+c+1)} \cdot \frac{\Gamma(a+1)\Gamma(b+c+1)}{\Gamma(a+b+c+2)}$$

$$= \frac{\Gamma(a+1)\Gamma(b+1)\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(a+b+c+2)}$$

$$= \frac{a\Gamma(a)\, b\Gamma(b)\, \Gamma(a+b+c)}{\Gamma(a)\Gamma(b) \cdot (a+b+c+1)(a+b+c)\Gamma(a+b+c)}$$

$$= \frac{ab}{(a+b+c)(a+b+c+1)}$$

$$Cov = E[XY] - E[X]E[Y]$$

$$= \frac{ab(a+b+c) - ab(a+b+c+1)}{(a+b+c)^2(a+b+c+1)}$$

$$= \frac{-ab}{(a+b+c)^2(a+b+c+1)}$$

$$Corr = \frac{\frac{-ab}{(a+b+c)^2(a+b+c+1)}}{\frac{\sqrt{ab(b+c)(a+c)}}{(a+b+c)^2(a+b+c+1)}}$$

$$= -\sqrt{\frac{ab}{(b+c)(a+c)}}$$

//

問 23

$$f_{1,\dots,k}(x_1,\dots,x_k \mid n,p_1,\dots,p_k)$$

$$= \frac{n!}{x_1!\cdots x_k!}\, p_1^{x_1}\cdots p_k^{x_k}$$

$$X_k \sim Bin(n, p_k)$$

(1)

$$f_{1,\dots,k-1|k} = \frac{f_{1,\dots,k}}{f_k}$$

$$= \frac{\frac{n!}{x_1!\cdots x_k!}\cdot p_1^{x_1}\cdots p_k^{x_k}}{\frac{n!}{x_k!(n-x_k)!}\, p_k^{x_k}(1-p_k)^{n-x_k}}$$

$$= \frac{(n-x_k)!}{x_1!\cdots x_{k-1}!}\, p_1^{x_1}\cdot \cdots \cdot p_{k-1}^{x_{k-1}}\cdot (1-p_k)^{-(n-x_k)}$$

$$= \frac{(n-x_k)!}{x_1!\cdots x_{k-1}!}\left(\frac{p_1}{1-p_k}\right)^{x_1}\cdot \cdots \cdot \left(\frac{p_{k-1}}{1-p_k}\right)^{x_{k-1}}$$

(2)

$$Var(X_i + X_j) = Var(X_i) + Var(X_j) + 2\,Cov(X_i, X_j)$$

$$X_i + X_j \sim Bin(n, p_i + p_j)$$
$$X_i \sim Bin(n, p_i)$$
$$X_j \sim Bin(n, p_j)$$ より

$$Cov(X_i, X_j) = \frac{Var(X_i + X_j) - Var(X_i) - Var(X_j)}{2}$$

$$= \frac{n(p_i+p_j)(1-p_i+p_j) - np_i(1-p_i) - np_j(1-p_j)}{2}$$

$$= \frac{\cancel{n(p_i+p_j)} - n(p_i+p_j)^2 - \cancel{np_i} - \cancel{np_j} + np_i^2 + np_j^2}{2}$$

$$= \frac{-2np_ip_j}{2} = \underline{-np_ip_j}$$

問24

$$X \sim N_k(\mu, \Sigma)$$

$t = (t_1, \cdots, t_k)^T$ のとき

$$M_X(t) = E[\exp\{t^T X\}]$$
$$= \exp\{\mu^T t + t^T \Sigma t/2\}$$ を示せ

$Z_1, \cdots, Z_k \sim N(0,1)$ i.i.d

$$Z = \begin{pmatrix} z_1 \\ \vdots \\ z_k \end{pmatrix}$$

$X = \Sigma^{\frac{1}{2}} Z + \mu$ とおく

$$M_X(t) = E[e^{t^T X}]$$
$$= E[e^{t^T \Sigma^{\frac{1}{2}} Z + t^T \mu}]$$
$$= e^{t^T \mu} E[e^{t^T \Sigma^{\frac{1}{2}} Z}]$$

$t^T \Sigma^{\frac{1}{2}} = s^T$ とおく

$$M_X(t) = e^{t^T \mu} E[e^{s^T Z}]$$
$$= e^{t^T \mu} E[e^{s_1 Z_1}] \cdots\cdots E[e^{s_k Z_k}]$$
$$= e^{t^T \mu} \cdot \prod_{i=1}^{k} \exp\left\{\frac{s_i^2}{2}\right\}$$
$$= e^{t^T \mu} \cdot \exp\left\{\sum_{i=1}^{k} \frac{s_i^2}{2}\right\}$$
$$= \exp\left\{t^T \mu + \frac{1}{2} s^T s\right\}$$

Σ は対称行列
$\Sigma^{\frac{1}{2}} \Sigma^{\frac{1}{2}T}$
$= \Sigma$

$$= \exp\left\{t^T \mu + \frac{1}{2}\left(t^T \Sigma^{\frac{1}{2}} \Sigma^{\frac{1}{2}T} t\right)\right\}$$
$$= \exp\left\{t^T \mu + \frac{t^T \Sigma t}{2}\right\}$$